5.3.4.Cell components and their functions

S.No	Cell Components	Main Functions	Special Name
1	Cell wall	• Surrounds and protects the cell • Make the cell stiff and strong	Supporter and protector
2	Cell membrane	• Holds and protects the cell • Controls the movement of materials in and out of the cell	Gate of the cell
3	Cytoplasm	• A watery, gel-like material in which cell parts move	Area of movement
4	Mitochondria	• Produce and supply most of the energy for the cell	Power house of the cell
5	Chloroplasts	• Contain green pigment chlorophyll • Capture the energy of sunlight and use it to produce food for the cell by photosynthesis.	Food producers for the cell (Plant cell)
6	Vacuoles	• Store food, water, and chemicals	Storage tanks
7	Nucleus	• Acts as 'brain' of the cell • Regulates and controls all the cell activities	Control centre
8	Nucleus membrane	• Surrounds and protects the nucleus control the movement of materials in and out of the nucleus	Gate of the nucleus

Points to Remember

- ❖ Cells are the basic units of all living organisms.
- ❖ There are two major cell types such as prokaryotic and eukaryotic cell.
- ❖ Both plant and animal cells have unique organelles which are capable of carrying out specialized functions.
- ❖ Plant cells have two unique components such as cell wall and chloroplasts compared to animal cells.

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